UNIT- III ACTIVITY PLANNING AND RISK MANAGEMENT

- Objectives of Activity planning
- Project schedules
- Activities
- Sequencing and scheduling
- Network Planning models
- Forward Pass & Backward Pass techniques
- Critical path (CRM) method
- Risk identification
- Assessment
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- Resource Allocation
- Creation of critical patterns
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Unit-III ACTIVITY PLANNING AND RISK MANAGEMENT.

Objectives of Activity planning.

- Activity planning is the most basic Process of Project and for individual activities the detailed project is enable to the following:

- Ensure that the appropriate resources will be available precisely when needed.
- Avoid different activities compares for the Same resource at Same time.
- Produce a detailed plan against which actual achievement may be measured.
- Produced a timed cash flow forecast.
- To replan the project during its life to Correct drift from the target.

- To be effective a plan must be stated as a target start and Completion date for each activity. Both are clearly visible.

- As a Project progress it is unlikely that everything Will go according to the plan.

Objectives of Activity planning:

- Project provides addition of progress and resource schedules activity planning aims to active a no. of related objectives.

- Some of the activities such as.

- Feasibility assessment : → The project feasibility assessment possible within required time scales and resource constraints.

- Resource allocation → The project plan allows us to investigate the relationship between timescales and resource availability and the efficacy of additional spending on resource procurement.

- Detailed costing → project cost and expenditure after produce activity plan and allocating specific resources, this is the estimation of cost and their timing.

- Motivation → It provide targets and being seen to monitor achievement against target effective way of motivating staff mainly where they have been involved in the first place targets.

- Coordination → The project plan particularly with large projects inviting more than a single project team provides an effective vehicle for communication and coordination among teams.

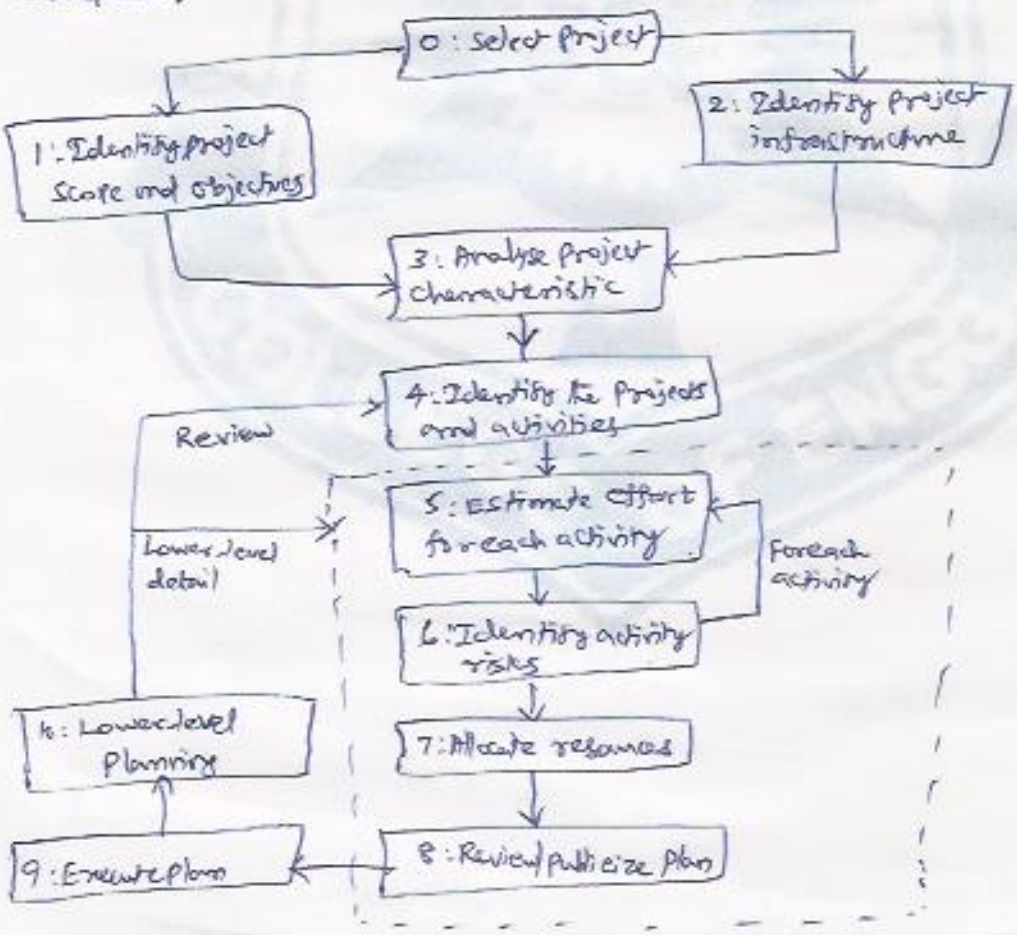
— Activity planning and scheduling techniques place an emphasis on completing the project in a minimum time at a acceptable cost or a minimum cost.

Project Schedules.

- Before the project schedule, project plan must be developed to the level of showing dates when each activity should start and finish and when and how much resource are needed. once the plan has been refined this level of detail, to create a project schedule comprises 4 mainstages:

Step-1: producing the plan is to decide what activities need to be carried out and in what order they are to be done.

Step-2: From this we can construct an "ideal activity plan". - that is, a plan of when each activity would ideally be undertaken were resources not a constraint. This activity plan is generated by step 4 and 5 of stepwise.



Activity planning is carried out in steps 4 and 5

Step 3: This step is resource allocation. The expected availability of resource might place constraints or when certain constraints activity can be carried out.

Final step: The final step is schedule production once resource have been allocated to each activity, it will be in a position to draw up and publish a project schedule, which indicates planned start and completion statements for each activity.

Projects - Activities.

① Defining Activities:

- Before the user try to identify the activities to add some assumptions that will be relevant when we start to produce an activity plan.

- A project is composed of a no. of interrelated activities.
- A project may start when at least one of its activities is ready to start.
- A project will be completed when all of the activities it encompasses have been completed.
- An activity must have a clearly defined start and a clearly defined end-point, normally marked by the production of a tangible deliverable.
- If an activity requires a resource then that resource requirement must be forecastable and is assumed to be required at a constant level throughout the duration of the activity.
- The duration of an activity must be forecastable - assuming normal circumstances and the reasonable availability of resources.
- Some activities might require that others are completed before they can begin (these are known as Precedence requirements).

② Identifying Activities:

- Essentially there are 3 approaches to identifying the activities or tasks that make up a project.

i) The activity-based approach
ii) The product-based approach
iii) The hybrid Approach.

i) The Activity-based Approach :

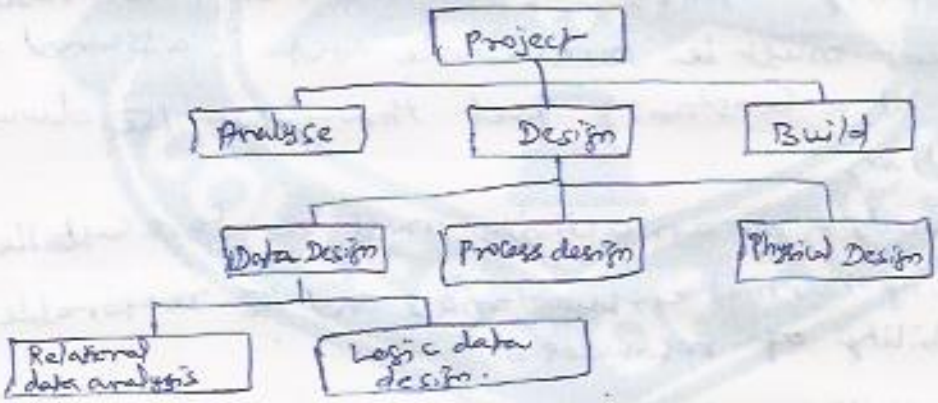
- It consists of creating a list of all the activities that the project is thought to involve.

- This might involve a brainstorming session involving the whole project team or it might stem from an analysis of similar past projects.

- When listing activities, particularly for a large project, it might be helpful to subdivide the project into the main life-style stages and consider each of these separately.

- Rather than doing this in an ad hoc manner, with the obvious risks of omitting or double-counting tasks, a much favoured way of generating a task list is to create a Work Breakdown Structure (WBS).

- WBS involves identifying the main (or high-level) tasks required to complete a project and then breaking each of these down into a set of lower-level tasks.



A fragment of an activity-based WBS.

Advantages :

- The WBS approach include the belief that it is much more likely to result in a task catalogue that is complete and is composed of non-overlapping activities.

- It is only the leaves of the structure that comprise the list of activities in the project - higher level nodes merely represent collection of activities.

ii) The Product-based Approach:

- It consists of producing a product Breakdown Structure (PBS) and a Product Flow Diagram (PFD).

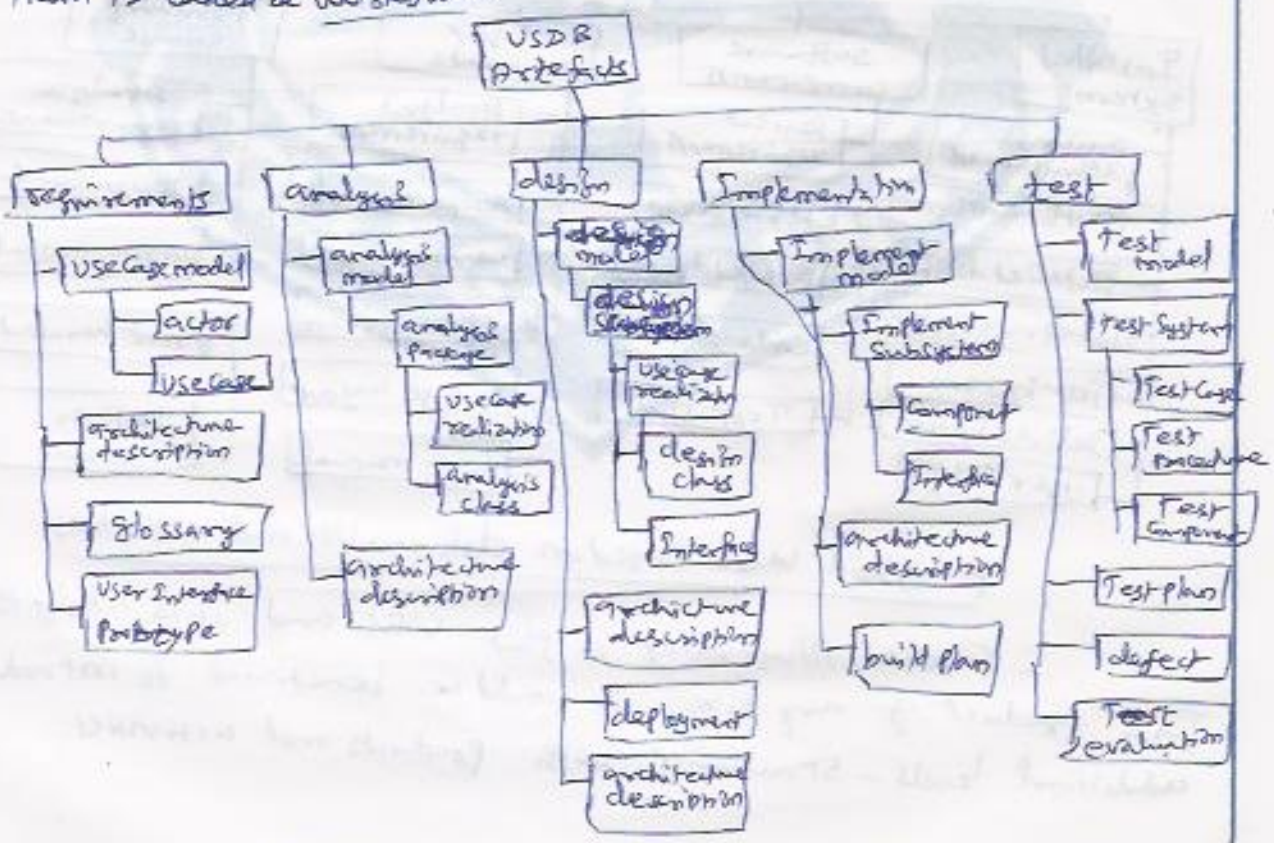
- The PFD indicates, for each product, which other products are required as inputs.

- The PFD can therefore be easily transformed into an ordered list of activities by identifying the transformations that turn products into others.

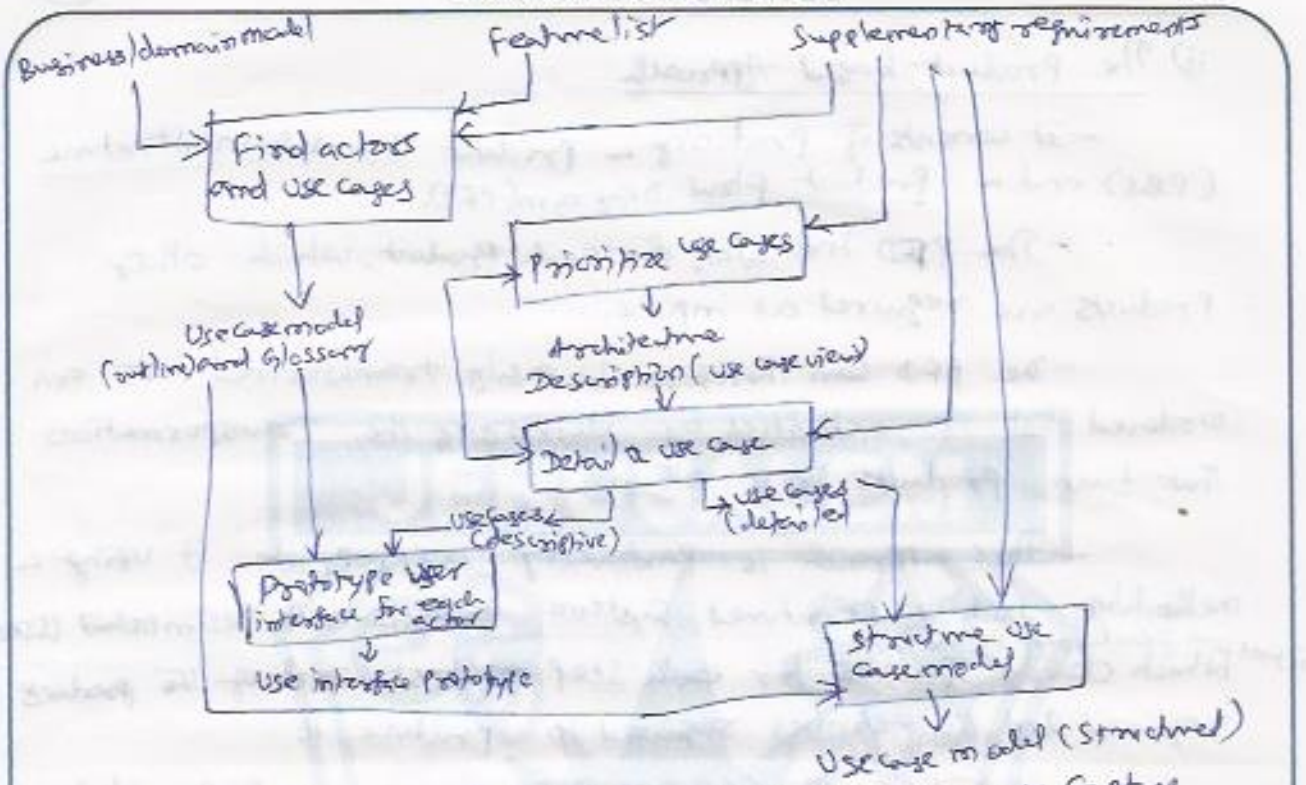
- This approach is particularly appropriate if using a methodology such as Structured Systems Analysis and Design Method (SSADM), or USDP (unified Software Development Process), which clearly specifies, for each step or task, each of the products required and the activities required to produce it.

- For eg, the SSADM Reference manual provides a set of generic PBSs for each stage in SSADM, which can be used as a basis for generating a project specific PBS.

- In the USDP, products are referred to as artifacts, see below figure, and the sequence of activities needed to create them is called a workflow.



USDP Product breakdown structure based on artefacts.



A structuring of activities for the USDB requirements capture workflow

iii) The Hybrid Approach:

A hybrid WBS based on deliverables and activities

- Figure illustrates a flat WBS and it is likely that in a product of any size, it would be beneficial to introduce additional levels - structuring both products and activities.

- A framework dictating the no. of levels and the nature of each level in the structure may be imposed on a WBS.

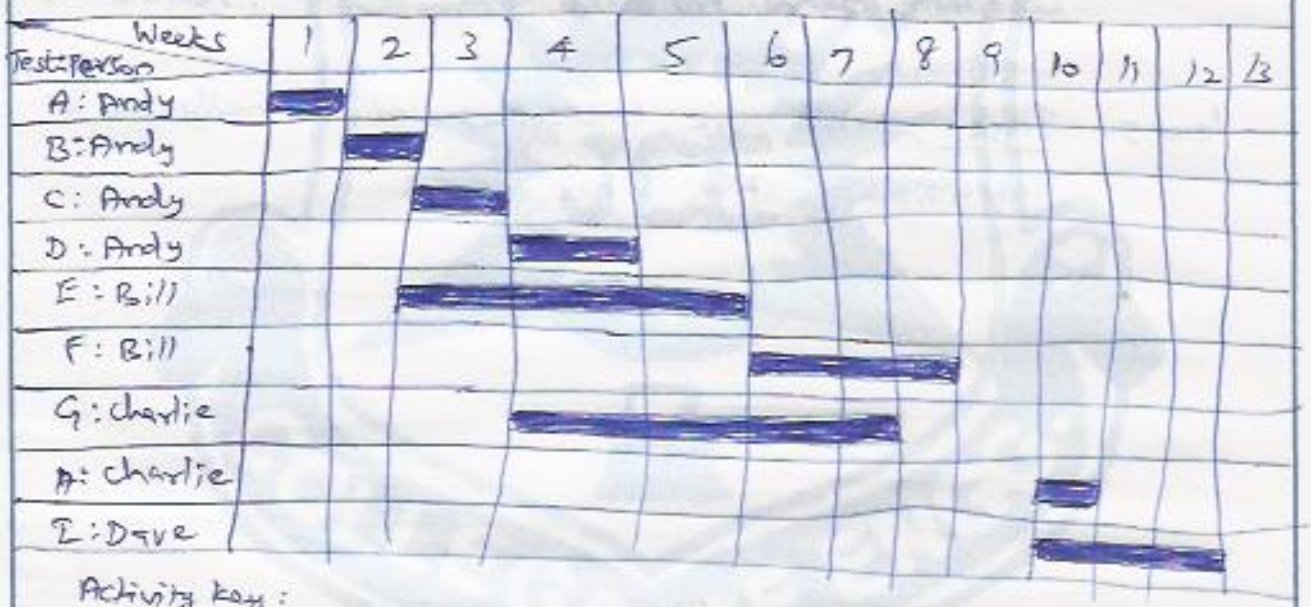
- For eg, in their MITP methodology, IBM recommend that the following 5 levels should be used in a WBS.

level 1 : Project
level 2 : Deliverables. such as sw, manuals and training courses.
level 3 : components, which are the key work items needed to produce deliverables, such as the modules and tests required to produce the system sw.
level 4 : Work-packages, which are major work items, or collections of related tasks, required to produce a component.
level 5 : Tasks, which are tasks that will normally be the responsibility of a single person.

Sequencing and scheduling.

– Throughout a project, we will require a schedule that clearly indicates when each of the project's activities is planned to occur and what resources it will need.

– The below chart has been drawn up taking account of the nature of the development process (that is, certain tasks must be completed before others may start) and the resources that are available (for example, activity C follows activity B because Andy cannot work on both tasks at the same time).



Activity key:

A : overall design
B : specify module 1
C : " 2
D : " 3
E : Code module 1
F : Code module 2
G : " 3
H : Integration testing
I : System testing

A project plan as a Bar Chart.

– In drawing up the chart, we have therefore done 2 things.

i) sequenced the tasks
ii) scheduled them.

Network Planning Models

– These project scheduling techniques model the project's activities and their relationships as a network. In the network, time flows from left to right.

– The two best known being techniques

- CPM (Critical path method)
- PERT (Program Evaluation Review Technique).

– Both of these techniques used an activity-on-arrow approach to visualizing the project as a network where activities are drawn as arrows joining circles, or nodes which represent the possible start and/or completion of an activity or set of activities.

– More recently a variation on these techniques, called precedence network, has become popular. This method uses activity-on-node networks where activities are represented as nodes and the links b/w nodes represent precedence requirements.

In the following, will look at the critical path method applied to precedence (activity-on-node) networks followed by a brief introduction to activity-on-arrow networks.

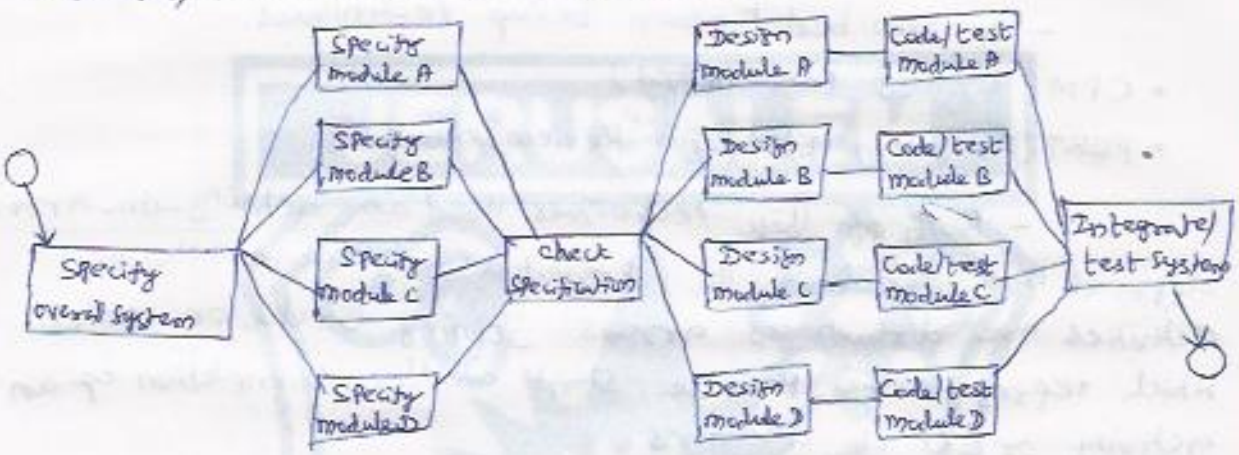
1) Formulating a Network model:

– The first stage in creating a network model is to represent the activities and their interrelationships as a graph

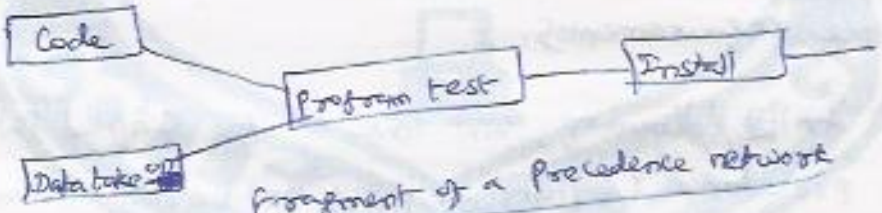
– In activity-on-node we do this by representing activities as links (arrowed lines) in the graph – the nodes (circles) representing the events of activities starting and finishing.

2) Constructing Precedence Networks:

- A project n/w should have only one start node.
- " " " end node.
- A node has duration – A node represents an activity, and in general, activities take time to execute.

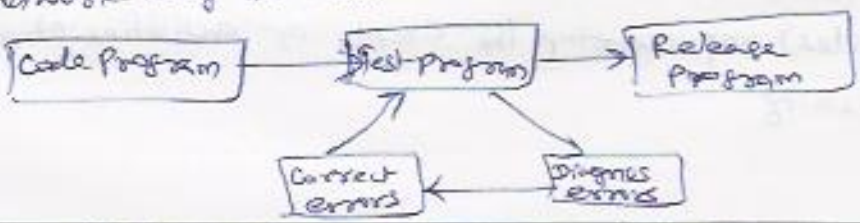


- Links normally have no duration. – Link represent the relationships b/w activities. In below figure, installation cannot start until program testing is complete. program testing cannot start until both coding and data take-on have been completed.



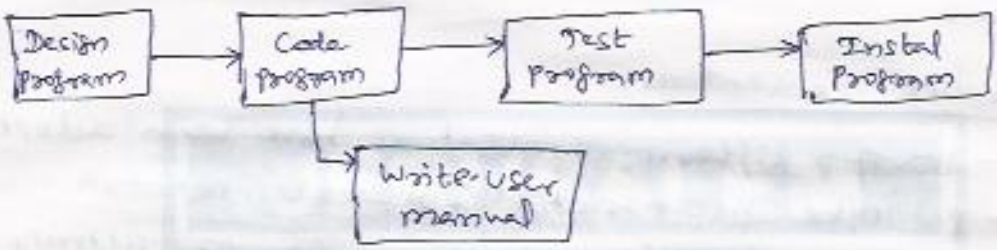
fragment of a precedence network

- Precedents are the immediate preceding activities. – In the above diagram, code and Data take-on are precedence of program test, program test is the precedence of Install.
- Time moves from left to right.
- A network may not contain loops.



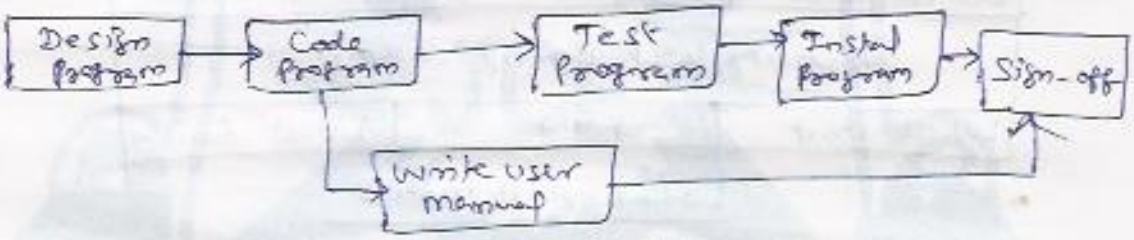
A loop represents an impossible sequence.

- A Network should not contain dangles – A dangling activity such as "write user manual" in below diagram should not exist as it is likely to lead to errors in subsequent analysis.



A dangle.

Project is complete once the s/w has been installed and the user manual written then we should redraw the n/w with a final completion activity.



Resolving the dangle.

(3) Representing Lagged Activities:

- This indicates the documenting amendments can start one day after the start of prototyping testing and will be completed 2 days after prototype testing is completed.

Indicating lags

(4) Hammock activities:

- are activities which in themselves have a zero duration but are assumed to start at the same time as the "hammocked" activity and to end at the same time as he last once.

(5) Labeling Conventions:

- No. of differing conventions have been adopted for entering infor. on an activity-on-node network.
- The activity label is usually identify the activity and may incorporate a project code.
- The activity description normally be a brief activity name such as "Test take-on module".

Earliest start	Duration	Earliest finish
Activity label, activity description		
Latest start	Float	Latest finish

Labeling Activities

(6) Adding the time dimension:

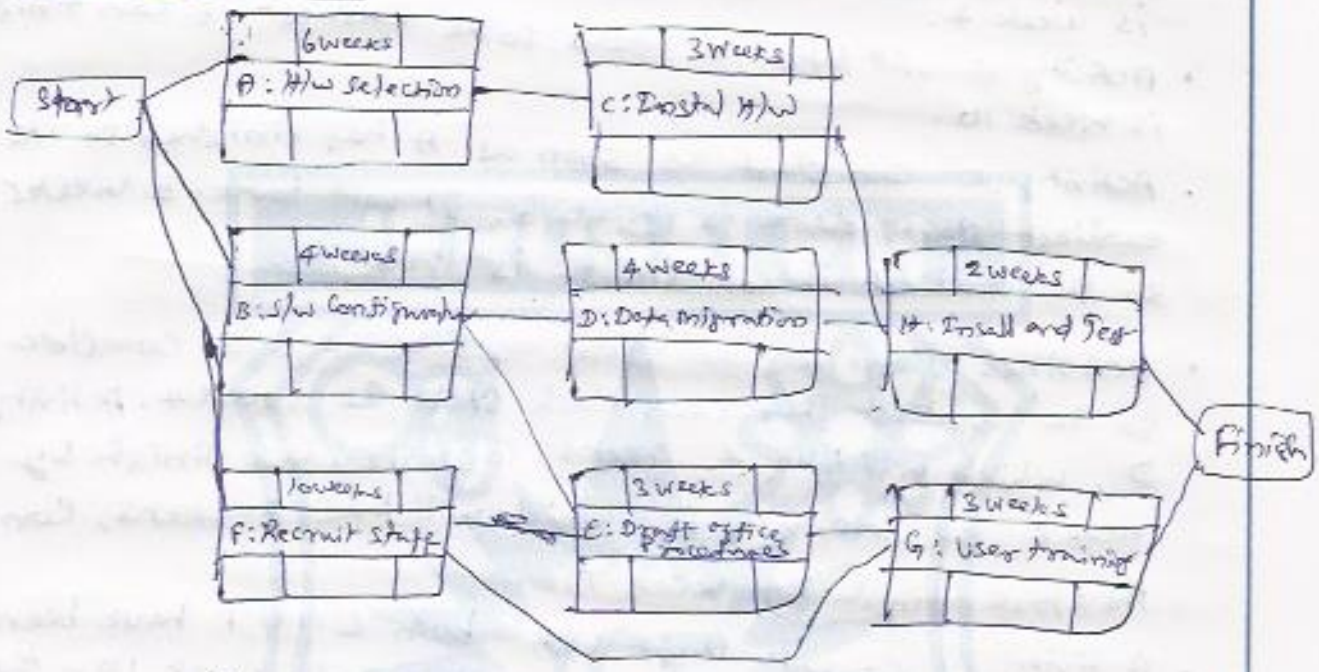
eg.

Activity		Duration (weeks)	Precedence
A	Hardware selection	6	
B	Software design	4	
C	Install hardware	3	A
D	Code and test software	4	B
E	File take-on	3	B
F	Write user manuals	10	
G	User training	3	E, F
H	Install & test system	2	C, D

Table: An example project specification with estimated activity durations and precedence requirements

Forward pass & Backward pass Techniques.

Forward pass:



The precedence n/w for the example project (refer previous page table)

- The forward pass is carried out to calculate the earliest dates on which each activity may be started and completed.

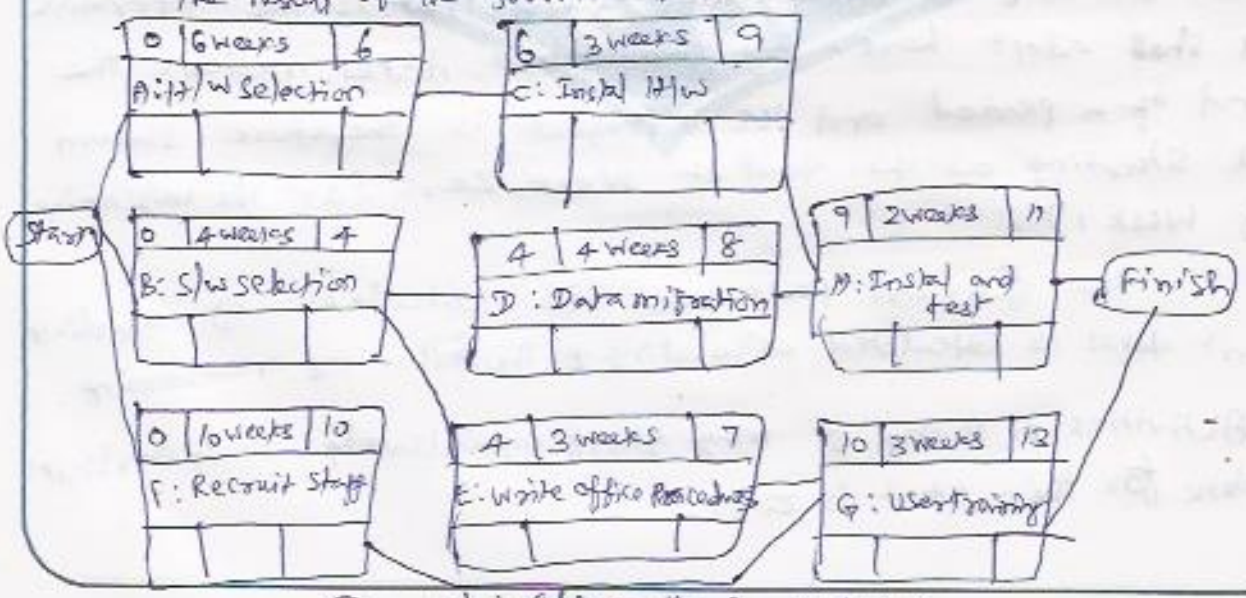
- When an actual start date is known, the calculations may be carried out using actual dates. Alternatively we can use day or week numbers and that is the approach we shall adopt here. By convention, dates indicate the end of a period and the project is therefore shown as starting at the end of week zero (or the beginning of week 1).

- The forward pass and the calculation of earliest start dates is calculated according to the following reasoning.

- Activities A, B and F may start immediately, so the earliest date for their start is zero.

- Activity A will take 6 weeks, so the earliest it can finish is week 6.
- Activity B will take 4 weeks, so the earliest it can finish is week 4.
- Activity F will take 10 weeks, so the earliest it can finish is week 10.
- Activity C can start as soon as A has finished so its earliest start date is week 6. It will take 3 weeks so the earliest it can finish is week 9.
- Activities D and E can start as soon as B is complete so the earliest they can each start is week 4. Activity D, which will take 4 weeks, can therefore finish by week 8 and activity E, which will take 3 weeks, can therefore finish by week 7.
- Activity G cannot start until both E and F have been completed. It cannot therefore start until week 10 – the later of weeks 7 (for activity E) and 10 (for activity F). It takes 3 weeks and finishes in week 13.
- Similarly, activity H cannot start until week 9 – the later of the two earliest finish dates for the preceding activities C and D.
- The project will be complete when both activities H and G have been completed. Thus the earliest project completion date will be the later of weeks 11 and 13 – that is, week 13.

The result of the forward pass are shown below.



The n/w after the forward pass

Backward pass:

– The second stage in the analysis of a Critical path n/w is to carry out a backward pass to calculate the latest date at which each activity may be started and finished without delaying the end date of the project.

– In calculating the latest dates, we assume that the latest finish date for the project is the same as the earliest finish date – that is we wish to complete the project as early as possible.

– The following shows the n/w after carrying the backward pass.

- The latest completion date for activities G and H is assumed to be week 13.
- Activity H must therefore start at week 11 at the latest (13 – 2) and the latest start date for activity G is week 10 (13 – 3).
- The latest completion date for activities C and D is the latest date at which activity H must start – that is, week 11. They therefore have latest start dates of week 8 (11 – 3) and week 7 (11 – 4) respectively.
- Activities E and F must be completed by week 10 so there earliest start dates are weeks 7 (10 – 3) and 0 (10 – 10) respectively.
- Activity B must be completed by week 7 (the latest start date for both activities D and E) so its latest start is week 3 (7 – 4).
- Activity A must be completed by week 8 (the latest start time for activity C) so its latest start is week 2 (8 – 6).
- The latest start date for the project start is the earliest of the latest start dates for activities A, B and F. This is week zero.

The n/w after the backward pass

Critical Path Method (CPM)

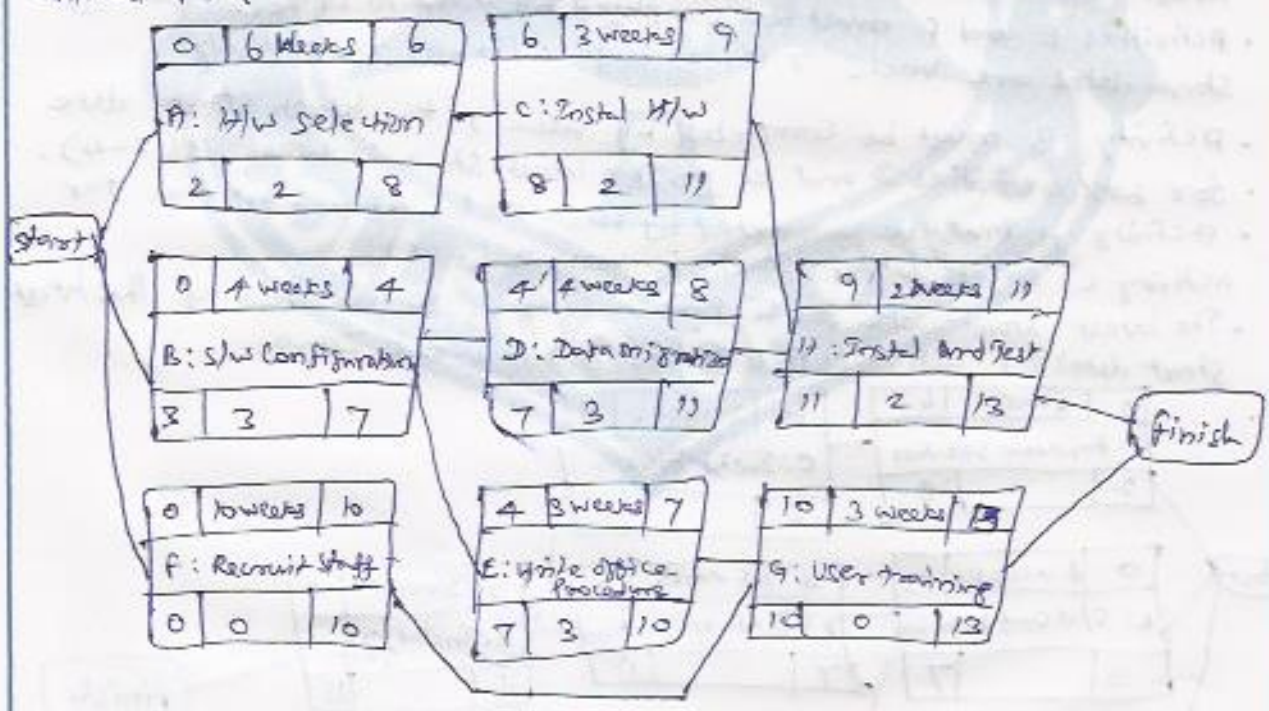
– There will be atleast one path through the n/w (that is, one set of successive activities) that defines the duration of the project. This is known as the Critical path.

– Any delay to any activity on this critical path will delay the completion of the project.

– The difference b/w an activity's earliest start date and its latest start date is known as the activity's float. – it is a measure of how much the start or completion of an activity may be delayed without affecting the end date of the project.

– Any activity with a float of zero is critical in the sense that any delay in carrying out the activity will delay the completion date of the project as a whole.

– There will always be atleast one path through the n/w joining these critical activities – this path is known as the critical path.



The Critical path

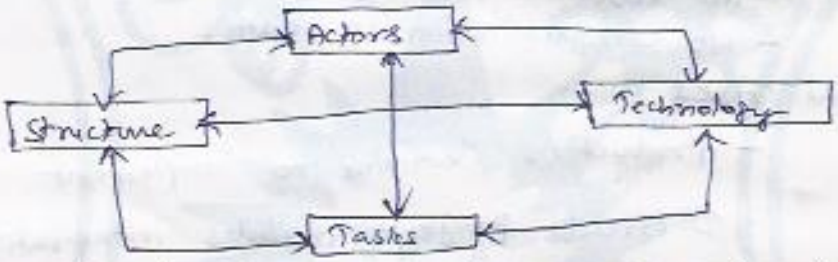
RISK MANAGEMENT

Risk:

– Every project involves risk. Risk is "an uncertain event or condition that, if it occurs has a positive or negative effect on a project objectives", include transferring the risk to another party, avoiding the risk, reducing the negative effect of the risk and accepting some or all of the consequences of a particular risk.

– 2 types of risks: Project risk, Business risk.

– Categories of risk:



- Actors → relate to all those involved in the project including both developers, users and managers.
  - Includes various department specialists, user groups, managers with different responsibilities.
  - if developer builds s/w components & leave before testing, team member taking over that component find lack of familarity with s/w make correction of faults difficult.
  - eg. a risk could be that high staff turnover leads to infor. of importance to the project being lost.
- Technology → encomposses both the technology used to implement the application and that embedded in the delivered products.
- Structure → This includes management procedures, risk here is that a group who need to carry out a particular project task are not informed of this need because they are not part of the project communication n/w.
- Tasks → The work to be carried out. A typical risk is that the amount of effort needed to carry out the task is underestimated.

Some other Categories of risks:

- Schedule risk.
  - wrong time estimation
  - Resources are not tracked properly.
  - Unexpected project scope expansions.
- Budget Risk
  - wrong budget estimation
  - Cost overruns
  - project scope expansion.
- operational Risks
  - NO proper subject training
  - NO resource planning
  - NO communication in team.
- Technical risks
  - Product is Complex to implement
  - Difficult project modules integration
  - Continuous changing requirements
- Programmatic risks
  - Market development
  - changing customer product strategy and priority
  - Government rule changes.

A Framework for Dealing With Risk.

The planning for risk includes these steps:

✓ Risk Identification – What risks might there be?

✓ Risk Analysis and Prioritization – Which are the most serious risks?

✓ Risk Planning – What are we going to do about them?

✓ Risk monitoring – What is the current state of the risk?

Risk Identification.

- The two main approaches to the identification of risks are the use of checklists and brainstorming.

- Checklists → are simply lists of the risks that have been found to occur regularly in s/w development projects. A specialized list of s/w development risks by Barry Boehm given below.

Risk	Risk reduction techniques
Personnel shortfalls	Staffing with top talent; job matching; teambuilding; training and career development; early scheduling of key personnel.
Unrealistic time and cost estimates.	Multiple estimation techniques; design to cost; incremental development; recording and analysis of past projects; standardization of methods.
Developing the wrong s/w functions.	Improved s/w evaluation; formal specification methods; user surveys; prototyping; early user manuals.
Developing the wrong user interface	Prototyping; task analysis; user involvement.
Gold plating	Requirements scrubbing; prototyping; cost-benefit analysis; design to cost.
Real-time performance shortfalls	Simulation; benchmarking; prototyping; tuning; technical analysis.
Development technically too difficult	Technical analysis; cost-benefit analysis; prototyping; staff training and development.

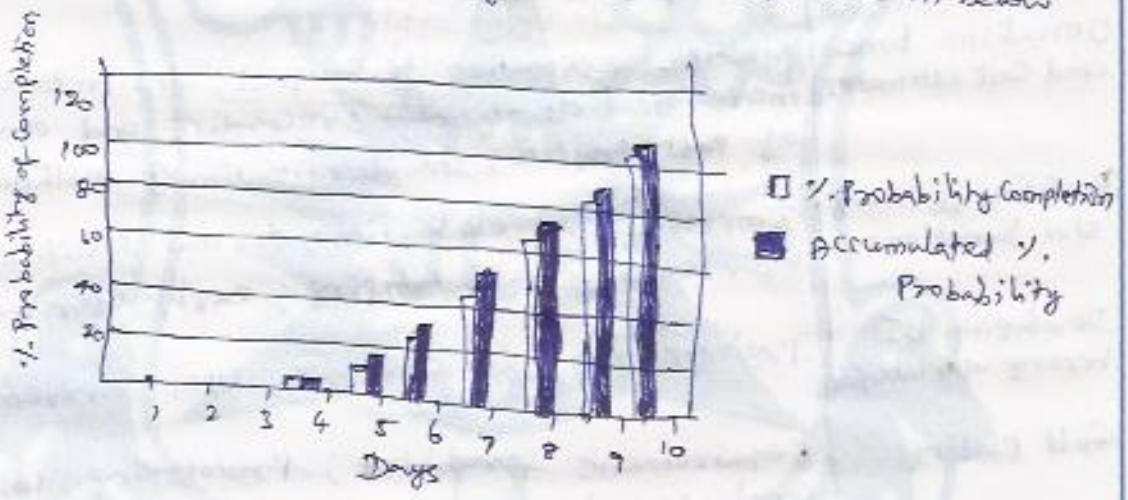
- Brainstorming → Representative of major stakeholders can be brought together and a plan is drafted.
  - It is used to identify the possible solutions to the problem.
  - All stakeholders have a meeting and risk in the projects are discussed.

Risk Assessment

- A common problem with risk identification is that a list of risks is potentially endless.
- A way is needed of distinguishing the damaging and likely risks. This can be done by estimating the risk exposure for each risk using the formula:

Risk exposure = (potential damage) x (probability of occurrence)

- The testing of a sw component is scheduled to take 6 days, but is actually done is 3 days. A team leader might therefore feel justified in producing a probability chart like the one in below figure.



Probability Chart.

Reference	Hazard	Likelihood	Impact	Risk
R1	Changes to requirements specification during coding	8	8	64
R2	Specification longer than expected	3	7	21
R3	Significant staff sickness affecting critical path activities	5	5	35
R4	Significant staff sickness affecting non critical activities	10	3	30
R5	Module coding takes longer than expected	4	5	20
R6	Module testing demonstrates errors or deficiencies in design	4	8	32

Part of Amanda's risk exposure assessment.

Impact level	Range
High	More than 30% above budgeted expenditure.
Significant	20 to 29% above budgeted expenditure.
Moderate	10 to 19% above budgeted expenditure
Low	Within 10% of budgeted expenditure.

Table: 1 Qualitative describtors of impact on cost and associated range values

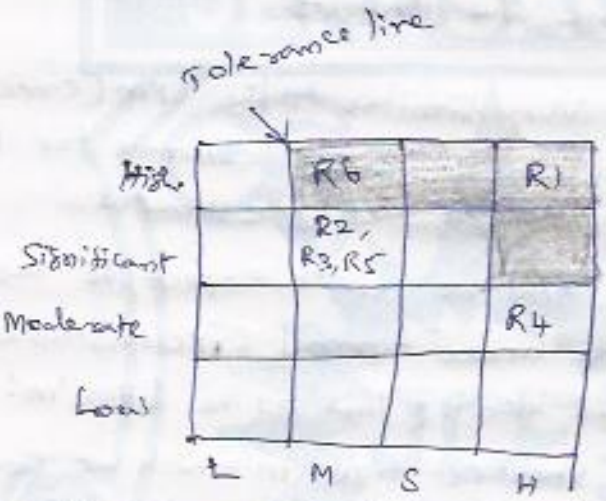


Fig: 1 A probability impact matrix

Table: 1 — The potential amount of damage has been categorized in terms of its impact on project costs. Other tables could show the impact of risks on project duration or on the quality of the project deliverables.

— In fig-1 some of the cells in the top right of the matrix have been zoned off by a tolerance line. Risks that appear within this zone have a degree of seriousness that calls for particular attention.

PERT Technique

- PERT (Program Evaluation and Review Technique) was developed to take account of the uncertainty surrounding estimates of task durations.

- It was developed in an environment of expensive, high-risk and state-of-the art projects - not that dissimilar to many of today's large s/w projects.

- The method is very similar to the CPM (Critical path method), but, instead of using a single estimate for the duration of each task, PERT requires 3 estimates.

- Most likely time → the time we would expect the task to take under normal circumstances. We shall identify this by the letter 'm'.
- Optimistic time → the shortest time in which we could expect to complete the activity, barring outright miracles. We shall use the letter 'a' for this.
- Pessimistic time → the worst possible time, allowing for all reasonable eventualities but excluding 'acts of God and warfare'. We shall call this 'b'.

PERT then combines these 3 estimates to form a single expected duration, t_e, using the formula

$$t_e = \frac{a + 4m + b}{6}.$$

Advantages:

- It seems by requesting multi valued activity duration estimates and calculate expected dates, PERT focuses attention of uncertainty of forecasting.
- So user can use the to rank them according to their degree of risk.
- Based on the expected times and standard deviations for forward passes through the n/w, for any event or activity completion, estimate the probability of meeting any set target.

Monte Carlo Simulation.

– An alternative to the PERT technique and to provide a greater degree of flexibility in specifying likely activity durations, user can use Monte Carlo Simulation techniques to evaluate risk of not achieving deadlines.

– When it is used to analyse the risk of not meeting the project deadline, the project completion time is first modelled as a mathematical expression involving the probability distributions of the completion times of various project activities and their precedence relationships.

– It essentially evaluates a range of input values generated from the specified probability distributions of the activity durations.

– It then calculates the results repeatedly; each time using a different set of random values generated from the given probability functions.

– After the simulation results are available, these are analysed, summarized and represented graphically, possibly using a histogram as shown below.

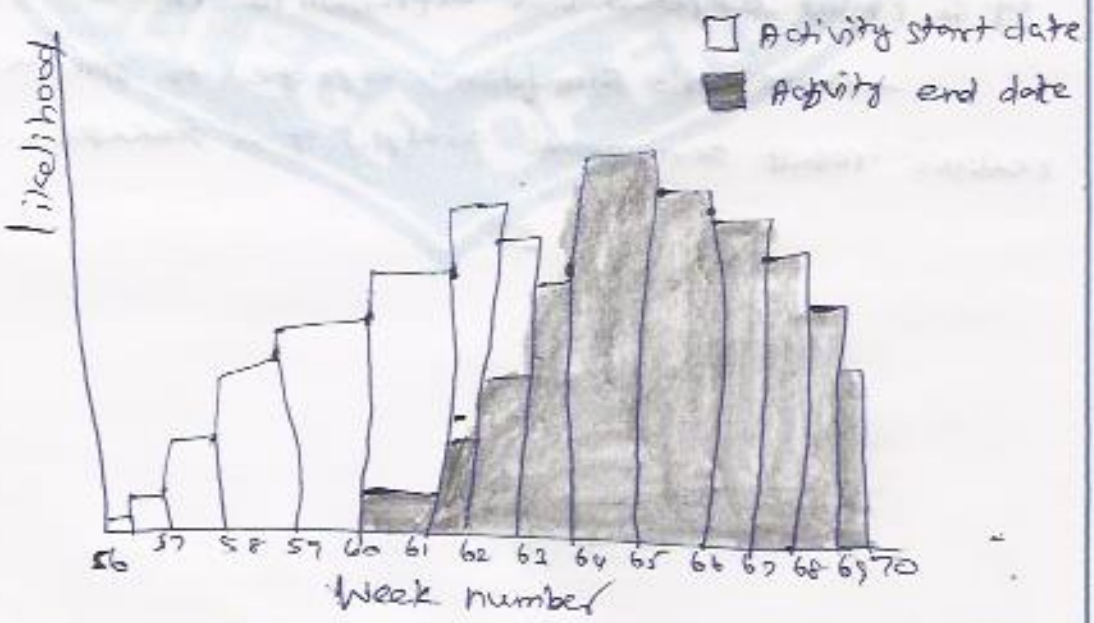


diagram-1 Risk profile for an activity generated using Monte Carlo Simulation

– The main steps involved in carrying out Monte Carlo simulation for a project consisting of n activities are as follows:

Step-1: Express the project completion time in terms of the duration of the n activities ($x_i, i=1,n$ and their dependences as a precedence graph, $d = f(x_1, x_2 .. x_n)$.

Step-2: Generate a set of random inputs, $x_{i1}, x_{i2} .. x_{in}$ using specified probability distributions.

Step-3: Evaluate the project completion time expression and store the result in d_i.

Step-4: Repeat steps 2 and 3 for the specified no. of times.

Step-5: Analyze the results $d_i, i=1,n$; summarize and display using a histogram as the one shown in previous page diagram (diagram-1).

Advantage:

– In monte carlo simulation, hundreds or thousands of possible random sampling of probability distribution functions of the activity duration are considered as samples for evaluation of the project completion time expression to produce outcomes.

– Monte carlo simulation is expected to give a more realistic result than manual analysis of a few cases.

RESOURCE ALLOCATION

- It describes about the how to match the activity plan to available resources and where necessary assess the efficacy of changing to plan fit the resources.

- Generally the allocation of resource to activities will lead us to review and modify the ideal activity plan. It may cause use to revise stage project completion dates.

- It is likely to lead a narrowing of the time spans with in which activities should be scheduled.

- The final result of resource allocation will normally be a no. of schedules include.

- Activity schedule → indicating the planned start and completion dates for each activity.
- Resource schedule → Showing the dates on which each resource will be required and the level of that requirement.
- Cost schedule → Showing the planned cumulative expenditure incurred by the use of resources over time.

The nature of Resources:

- A resource is any item or person required for execution of the project.

- Some of the resource such as project manager required for the direction of project whereas others, such as specific s/w developer might be required for single activity.

- Generally resources will fall into one of 7 categories

1. Labour → The main items in this category will be members of the development project team such as the project manager, systems analysts and s/w developers.
2. Equipment → Obvious items will include workstations and other computing and office equipment.
3. Materials → are items that are consumed, rather than equipment that is used.

4. Space → For projects that are undertaken with existing staff space is normally readily available. If any additional staff (recruited (or) contracted) should be needed then office space will need to be found.

5. Services → Some projects will require procurement of specialist services - development of a wide area distributed system, for ex, requires scheduling of telecommunication services.

6. Time → is the resource that is being offset against the other primary resources - project timescales can sometimes be reduced by increasing other resources and will almost certainly be extended if they are unexpectedly reduced.

7. Money → is a secondary resource - it is used to buy other resources and will be consumed as other resources are used.

Identifying resource requirements:

- First step in producing a resource allocation plan is to list the resource will be required along with the expected level of demand.

- It is likely however there will be also be resources required are not activity specific but are not part of project's infrastructure or required to support other resources

Sheduling resources.

- It is to be produced the resource requirements list, the next stage is to map this on the activity plan to assess the distribution of resources required over the duration of the project.

- The following diagram shows Amanda's activity plan as a bar chart and a resource histogram for analyst/designers. Each activity has been scheduled to start at its earliest start date a sensible initial strategy.

Part of Amanda's bar chart and resource histogram for analyst/designers

- Resources have to be allocated to a project on an activity-by-activity basis and finding the 'best' allocation can be time consuming and difficult.

- Allocating a resource to one activity limits the flexibility for resource allocation and scheduling of other activities.

- Prioritize activities can be allocated to competing activities in some rational order.

- Various ways of prioritizing activities, two are given below.

- Total float Priority → Activities are ordered according to their total float, those with the smallest total float having the highest priority.

- Ordered list Priority → activities that can proceed at the same time are ordered according to a set of simple criteria.
  For eg, Burman's Priority list,

1. Shortest critical activity
2. critical activities
3. Shortest non-critical activity
4. non-critical activity with least float
5. non-critical activities.

Creating Critical paths.

- Scheduling paths resources can create new critical paths. Delaying the start of an activity to become critical uses this float.

- Amanda's revised schedule still calls for four analysts/designers but only for a single day, is illustrated in the solution. Delaying the specification of module c has used up all of its float and the subsequence activity along the path.

- It has now 2 critical paths:

  1) One shows the precedence network
  2) The new one.

- In a large project resource linked criticalities can be quite complex a hint of the potential problem may be appreciated by look at the next exercise.

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Cost Schedules.

– This scheduled process showing the weekly or monthly costs over the life of the project.

– Calculating the cost is straight forward where the organization has standard cost figures for staff and resources.

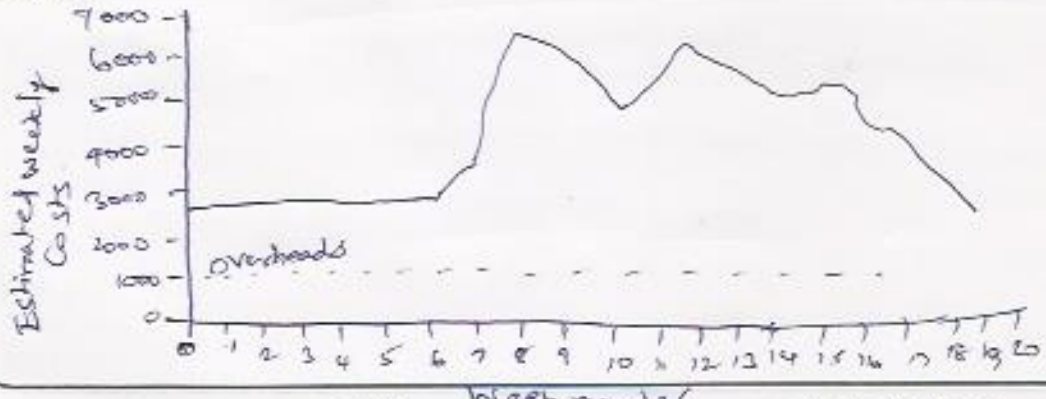
– In general, costs are categorized as follows:

- Staff costs → These will include staff salaries as well as the other direct costs of employment such as the employer's contribution to social security funds, pension scheme contributions, holiday pay and sickness benefit.

- Overheads → It represents expenditure that an organization incurs, which cannot be directly related to individual projects or jobs, including space rental, interest charges and the costs of service departments (such as human resources).

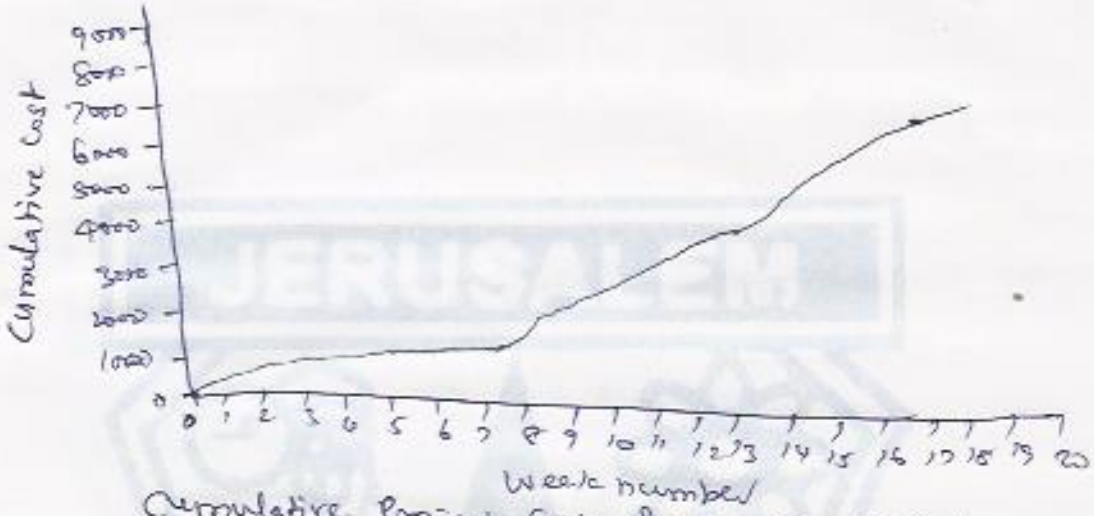
- Usage charges → In some organizations, projects are charged directly for use of resources such as computer time. This will normally be on an 'as used' basis.

– The following diagram shows the weekly costs over the 20 weeks that Amanda expects the project to take.



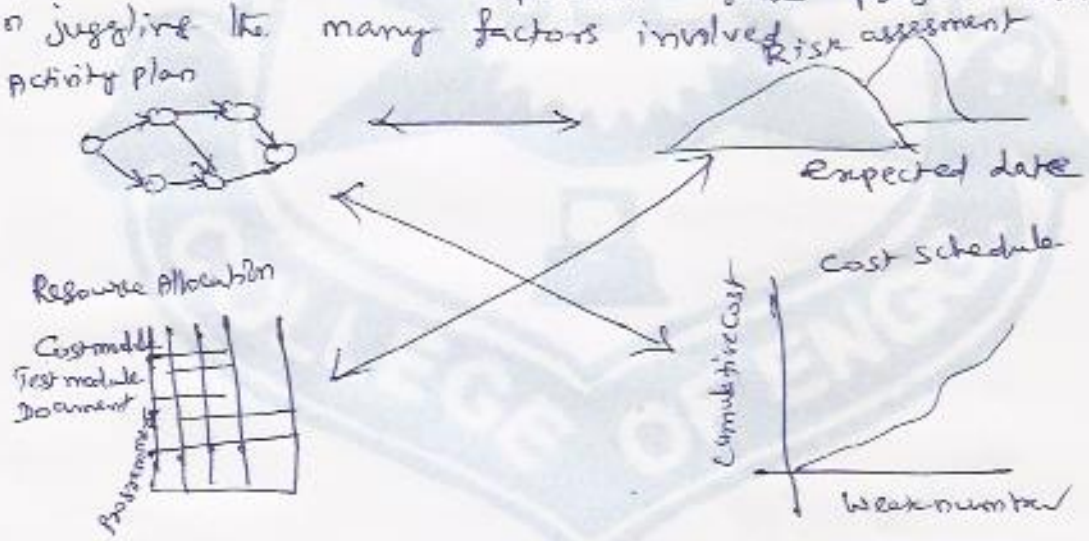
Weekly project costs for the JOE project

- The following diagram shows the cumulative cost of the project and it is generally this that would be used for cost control purposes.



Cumulative Project Costs for the IOE project

- Successful project scheduling is largely dependent upon the skill and experience of the project manager in juggling the many factors involved.



Successful project scheduling is not a simple sequence